

ERNESTO L M RODRIGUES FILHO

Software Engineer | Embedded Systems | Distributed & Cloud Infrastructure

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TECHNICAL SKILLS

Languages: C++, C, Java, Python, Bash, JavaScript/TypeScript, HTML, CSS, Kotlin

Systems & Infrastructure: Linux, FreeRTOS, Azure, Red Hat OpenShift, Docker, Kubernetes, GitHub Actions

Tools, Frameworks & Practices: CMake, Git, Spring Boot, REST, Snowflake, Datadog, k6, CI/CD, TDD, automated testing

EXPERIENCE

Software Engineer

Jan 2023 – Present

ExxonMobil Global Business Center Brasil

Curitiba, PR, Brazil

- Co-developed a centralized, configurable data-access service on Azure Red Hat OpenShift (ARO) that enables clients to deploy customized API endpoints through low-code YAML configurations in GitHub, providing self-service access to Snowflake data without requiring them to build and maintain their own microservices.
- Engineered the supporting CI/CD automation using GitHub Actions, Python, and Bash to validate configurations and streamline deployment of customer-specific service instances.
- Led improvements to production observability by designing custom telemetry, Datadog monitors, and Snowflake dashboards, providing real-time visibility into pipeline health and operational failures.
- Co-developed automated evaluation tooling for dbt projects and integrated it into pull-request workflows, strengthening pre-merge validation and preventing issues from reaching downstream environments.
- Spearheaded cross-platform build standardization across Windows, MacOS and Linux, eliminating the need for additional hardware and avoiding **\$120k+** in costs.
- Delivered 5+ internal web applications with Storybook library to ensure consistent user experience and responsive design, and architected a custom JavaScript interface to facilitate seamless data interactions with SharePoint's REST API.

Teaching Assistant – Fundamentals of Programming

Feb 2018 – Dec 2019

Universidade Tecnológica Federal do Paraná (UTFPR)

Curitiba, PR, Brazil

- Hosted and maintained an educational portal to assist students with compiler setup, library configuration, and code debugging.
- Provided technical guidance and troubleshooting support for undergraduate C/C++ programming coursework.

PROJECTS & OPEN SOURCE

Modular Monitor Project

Capstone Project

Embedded C++ | ESP32 | FreeRTOS | I²C | Multithreading | Device Drivers

github.com/ModularMonitor

- Developed a modular real-time monitoring engine for ESP32 with hot-swappable I²C sensor architectures, multithreading, and custom device drivers.
- Integrated TFT touchscreen UI rendering, SD-card logging, 4G cellular connectivity, and TCP wireless data streaming to a web interface.

Lunaris Library Collection

Open Source

Modern C++ (C++17/C++20/C++23) | Concurrency | Networking | Systems Programming

github.com/LunarisLib

- Authored a cross-platform utility library featuring lock-free thread primitives, asynchronous I/O, socket abstraction, encryption, and a memory-optimized JSON parser based on cJSON.
- Engineered a suite of 14 modules to accelerate application development by providing standardized, reusable C++ components. Implemented robust build and testing pipelines using CMake and its CTest runner tool.

Personal website

Open Source

HTML | CSS | JavaScript | React.js

github.com/Lohkat

- Developed and deployed a responsive web portfolio hosted on [GitHub Pages](#).
- Built interactive frontend features – dynamic galleries, web-player, and search tools – showcasing cross-domain technical adaptability.

EDUCATION

M.S. in Electrical Engineering & Industrial Informatics (CPGEI)

Jan 2026 – Present

Universidade Tecnológica Federal do Paraná (UTFPR)

Curitiba, PR, Brazil

- Specialization in Real-Time Embedded Systems, Signal Processing, and Industrial Automation.

B.S. in Electrical and Electronics Engineering

Jan 2016 – Oct 2024

Universidade Tecnológica Federal do Paraná (UTFPR)

Curitiba, PR, Brazil

LANGUAGES

Languages: Portuguese – Native, English – Advanced / Professional Working Proficiency